

DRIFT OR MIGRATION?

Testing SignalRupture Drift Against Rasmussen's Dynamic Risk Model

SignalRupture (SR)

Adversarial Comparative Theory Test

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Abstract

Drift is one of the central concepts in SignalRupture (SR). Yet safety science already contains a mature theory of gradual migration toward unsafe operating boundaries, most notably Jens Rasmussen's 1997 dynamic risk model and subsequent work on drift into failure. This paper attempts to break SR's drift claim by comparing it against Rasmussen's theory at the level of causal structure, system scale, observability, human adaptation, boundary behavior, propagation, and intervention. The comparison finds substantial overlap. Rasmussen already models socio-technical risk across multiple organizational levels, explicitly situates systems under technological, economic, regulatory, and public pressures, and explains how normal local adaptations can systematically migrate activity toward a boundary of acceptable performance. SR therefore cannot claim novelty merely for gradual deterioration, multi-level socio-technical drift, hidden erosion under successful operation, or pressure-driven migration toward failure boundaries. Several strong SR claims are narrowed or rejected. What survives is a more specific candidate contribution: SR treats drift as a measurable cross-domain state deviation that can occur in epistemic, infrastructural, diagnostic, and human-capacity layers, couples it to visibility lag and cross-layer propagation, and links drift diagnosis to restoration sufficiency. These additions are conceptually distinct from Rasmussen's safety-boundary migration, but their incremental predictive value is not yet established. The paper concludes that Rasmussen substantially anticipates SR's generic drift mechanism; SR survives only if its broader measurement and propagation architecture predicts or diagnoses structural deterioration beyond what dynamic safety migration already explains.

Keywords: drift; Rasmussen; dynamic risk; migration to the boundary; SignalRupture; safety science; sociotechnical systems; visibility lag; propagation; restoration

1. Research Question

The central question is deliberately destructive:

Is SR Drift a distinct construct, or is it Rasmussen-style migration toward a boundary under new terminology?

This is a critical test because drift appears throughout the SR architecture. If the underlying causal idea is already captured by established safety science, SR must acknowledge that provenance and restrict its novelty claims.

The comparison therefore gives Rasmussen the strongest possible interpretation rather than treating his work as a narrow accident model.

2. Rasmussen's Dynamic Risk Model

Rasmussen's 1997 model was developed for a dynamic socio-technical society in which risk management spans legislators, regulators, managers, work planners, and operators. He explicitly identifies fast technological change, competitive pressure, changing regulation, and public pressure as forces acting on the system (Rasmussen, 1997).

Human and organizational activity occurs within a space bounded by economic, workload, administrative, functional, and safety constraints. Local actors adapt continually. Because local adaptation is shaped by pressures for efficiency and least effort, successful excursions that do not immediately cause failure can normalize progressively riskier operation.

Local adaptation + efficiency pressure + weak/late safety feedback → migration toward boundary

Morrison and Wears later formalized Rasmussen's conceptual model using system dynamics and reproduced the reinforcing mechanism by which excursions can gradually shift the perceived acceptable boundary and normal operating point toward greater risk (Morrison & Wears, 2022).

ADVERSARIAL TEST — Do not understate Rasmussen

Rasmussen is not merely saying that workers make mistakes. His model is explicitly systemic, multi-level, dynamic, pressure-sensitive, and focused on how normal

adaptation can generate unsafe migration without malicious intent or obvious local failure.

3. SignalRupture Drift

SR uses drift as a general structural deviation from an earlier or reference state. In its simplest formal expression:

$$\text{Drift}(t) = |M_t - M_0|$$

The construct is used across multiple SR layers, including model drift, epistemic drift, infrastructural drift, institutional drift, and diagnostic drift. SR also links drift to delayed recognition and propagation:

$$\text{Drift} \uparrow + \text{VisibilityLag} \uparrow + \varphi \uparrow \rightarrow \text{rising structural instability}$$

In SR, drift does not require an accident boundary. A system can remain formally operational while its state moves away from a stable, coherent, or restorable configuration.

The strongest SR claim would therefore be that drift provides a broader and more measurable account of structural deterioration than migration-to-boundary theories.

4. Direct Conceptual Mapping

SR concept	Rasmussen analogue	Overlap	Possible SR remainder
Drift	Migration of operating point / boundary	Strong	SR applies deviation beyond safety-performance boundaries
Efficiency pressure	Cost/efficiency gradient	Very strong	SR may include broader infrastructural and epistemic pressures
Human compensation	Local adaptive search / effort gradient	Strong	SR explicitly models human capacity depletion in some domains
Visibility lag	Delayed/ambiguous safety feedback	Strong precursor	SR attempts to operationalize delay as a separate variable

Boundary crossing	Functionally acceptable performance boundary	Very strong	SR uses multiple collapse/restoration boundaries
Cross-level system	Legislator-to-operator socio-technical hierarchy	Very strong	SR emphasizes cross-domain coupling among infrastructures
Propagation ϕ	No direct single scalar equivalent	Partial	Candidate SR distinction
Restoration deficit	Not central to Rasmussen model	Limited	Candidate SR distinction

5. First Adversarial Result: Generic Drift Is Not Novel

The strongest immediate result is unfavorable to SR. The generic mechanism:

Pressure → adaptation → normalization → gradual movement toward failure

is already present in Rasmussen's theory.

So is the idea that locally rational behavior can produce global risk, that feedback about safety can be delayed or ambiguous, and that systems can operate successfully for long periods while moving toward a dangerous boundary.

SR therefore should not claim:

- that it uniquely discovered gradual migration toward failure under normal operation;
- that it uniquely discovered multi-level socio-technical drift;
- that it uniquely discovered pressure-driven adaptation as a source of systemic risk;
- that it uniquely discovered hidden erosion under apparent operational success.

ADVERSARIAL TEST — Terminological novelty is insufficient

Renaming migration as drift does not create a new scientific construct if the variables, causal relations, and predictions are functionally identical.

6. Second Result: Visibility Lag Has a Strong Rasmussen Precursor

SR often treats Visibility Lag as a distinct diagnostic variable: the interval between structural onset and institutional recognition.

$$\mathbf{VL = T_recognition - T_structural_change}$$

Rasmussen's mechanism already depends heavily on delayed or weak feedback. Protective layers can be violated without immediate visible consequence, and successful excursions teach the system that more aggressive operation appears acceptable. Morrison and Wears emphasize the asymmetric visibility of efficiency gains versus safety maintenance: performance improvement is immediate and visible, while safety feedback is delayed, ambiguous, or absent.

Therefore SR cannot claim conceptual novelty for delayed recognition itself.

The remaining SR question is whether explicitly measuring VL as a time variable improves diagnosis or prediction across domains.

Possible SR contribution = operationalized recognition delay, not discovery of delayed feedback

ADVERSARIAL TEST — Measurement burden

If VL is merely a relabeling of known feedback delay and adds no independent predictive value, it should be treated as an SR operationalization of an established safety-science mechanism rather than a novel law.

7. Third Result: Human Compensation Also Overlaps

Rasmussen's model is built around adaptive human and organizational behavior. Workers and managers continuously modify strategies under changing workload, cost, and safety constraints.

SR's Eroded Subject and compensated-continuity architecture adds a stronger claim:

InstitutionalDeficit → HumanCompensation → ObservedContinuity → possible HumanSlack loss

Only the last term moves materially beyond Rasmussen. Rasmussen explains how adaptation migrates operation toward a safety boundary; SR additionally asks whether the adaptive mechanism itself consumes the human substrate required for future restoration.

The present evidence supports human-compensation pressure as plausible but does not establish that every adaptive response is depleting. The resilience challenge already forced SR to distinguish renewable, stable, and depleting compensation.

ADVERSARIAL TEST — SR cannot appropriate adaptation

Human adjustment under pressure is deeply established in safety science and resilience engineering. SR's distinctive claim, if any, is not adaptation itself but measurable depletion of future restoration capacity caused by some forms of persistent compensation.

8. Where SR May Actually Differ: Domain of the Boundary

Rasmussen's central object is migration toward a boundary of functionally acceptable or safe performance. SR's drift construct is broader in domain. It can describe movement away from a reference state even where there is no immediate safety boundary:

- epistemic coherence can drift;
- model behavior can drift;
- institutional capacity can drift;
- diagnostic visibility can drift;
- housing affordability can drift;
- restoration capacity can drift.

This difference is real at the level of scope. But broader scope is not automatically greater explanatory power.

Broader applicability ≠ stronger theory

To earn distinct status, SR must show that the same drift formalism works across those domains without becoming so abstract that it merely means 'change over time.'

ADVERSARIAL TEST — Overgeneralization risk

Drift becomes vacuous if every deviation from a baseline counts. SR therefore requires domain-specific state variables, directional interpretation, thresholds, and falsification criteria. Otherwise Rasmussen's boundary model is scientifically tighter.

9. Propagation as the Strongest Candidate Distinction

SR's most promising difference is not drift itself but what happens after drift emerges. SR explicitly models cross-layer transfer using a propagation term ϕ .

ϕ = cross-layer drift-transfer / propagation potential

Rasmussen's system is multi-level and dynamic, but the classic migration model does not reduce cross-domain propagation to a single explicit diagnostic operator. SR attempts to ask:

Where does drift travel next, through which dependencies, and with what amplification?

For example, a staffing deficit may propagate into wait times, household burden, labor retention, public spending, and political pressure. An electricity constraint may propagate into data-centre deployment, institutional digital capacity, and service continuity.

This is a plausible SR extension, but it remains to be established that ϕ is measurable, stable, and superior to existing network, interdependency, or systems-dynamics methods.

ADVERSARIAL TEST — Network-science criticism

Propagation is not an untouched domain. Network science, cascading-failure research, systems dynamics, reliability engineering, and infrastructure interdependency models already study transfer and amplification. SR must compare ϕ against those methods rather than claim propagation itself as new.

10. Restoration as the Second Candidate Distinction

Rasmussen's theory explains how systems migrate toward dangerous boundaries. SRD asks an additional question:

Can the system rebuild durable capacity faster than the restoration requirement changes?

This gives SR a potentially different endpoint. Rasmussen is primarily an accident/risk-control theory; SRD is a restoration-sufficiency theory.

Rasmussen: migration risk → safety boundary

SRD: restoration requirement ↔ durable capacity ↔ slack ↔ temporal window

That distinction survived the resilience challenge in a limited form. Yet it belongs to SRD rather than generic SR drift. SR should therefore avoid using restoration as retrospective evidence that drift itself was novel.

11. Formal Causal Comparison

A simplified Rasmussen model can be represented conceptually as:

EfficiencyPressure_t + LeastEffort_t → OperatingPointMigration_t
SuccessfulExcursion_t → PerceivedBoundaryShift_{t+1} →
SafetyPressure_↓

A simplified SR model is:

Pressure_t → Drift_t
Drift_t + VisibilityLag_t + Coupling_t → Propagation_t
Propagation_t + LowSlack_t → BoundaryRisk_{t+1}

The first equation is heavily anticipated by Rasmussen. The candidate SR additions are the explicit coupling of drift to visibility, propagation, slack, and restoration.

The decisive empirical question becomes:

Does {VL, φ, Slack, RD} add predictive value after Rasmussen-style pressure/migration variables are included?

12. Empirical Strategy to Try to Break SR Further

A fair test requires a domain with repeated operational measurements and identifiable safety or functional boundaries. Candidate domains include hospital operations, grid adequacy, aviation maintenance, cloud infrastructure, and industrial process safety.

Three nested models should be compared:

M₀ = demand + baseline capacity + trend
M_{Rasmussen} = M₀ + efficiency pressure + workload pressure +
boundary proximity + adaptive excursions

$$\mathbf{M_SR = M_Rasmussen + Drift + VL + \phi + Slack/Restoration\ variables}$$

Outcomes should be future boundary violations, service deterioration, cascading impact, or restoration delay.

If $\text{Performance}(\mathbf{M_SR}) \leq \text{Performance}(\mathbf{M_Rasmussen})$, SR drift adds no demonstrated incremental value

ADVERSARIAL TEST — Complexity penalty

SR contains more variables. It must be evaluated out of sample and penalized for model complexity. Better historical fit alone is not evidence.

13. A Harder Problem for SR: Rasmussen Is Already Cross-Level

One tempting SR defense would be that Rasmussen only models local safety behavior while SR connects institutions, infrastructure, governance, and human systems. That defense does not survive the literature.

Rasmussen explicitly frames risk management as a control problem spanning government policy, regulators, company management, work planning, staff, and hazardous process. His 1997 argument was motivated by the inadequacy of studying each level independently.

Therefore SR cannot claim cross-level integration as a categorical difference.

The narrower candidate distinction is cross-domain coupling among heterogeneous social, digital, epistemic, and infrastructural systems rather than hierarchy within one hazard-control structure.

ADVERSARIAL TEST — Important concession

This is one of the strongest overlaps found. Any SR paper describing Rasmussen as operator-level safety theory would be inaccurate.

14. Another Problem for SR: 'Drift Into Failure' Is Already a Mature Research Program

Subsequent safety literature has applied and debated drift into failure across road transport, healthcare, industrial safety, and other complex systems. The concept has also been criticized for overgeneralization and normative assumptions.

That matters because SR faces the same vulnerability: a broad drift vocabulary can become an after-the-fact explanation for any deterioration.

SR therefore needs stronger prospective constraints than 'drift was present before failure.' It should specify:

- the state variable expected to drift;
- the direction of drift;
- the baseline and boundary;
- the expected propagation path;
- the recognition delay;
- the time window;
- what observation would count against the prediction.

Without those elements, SR drift risks becoming precisely the kind of broad explanatory construct that critics of drift-into-failure have challenged.

15. What Rasmussen Forces SR to Remove

- Generic gradual migration toward failure as an SR novelty claim.
- Pressure-driven adaptation as an SR novelty claim.
- Multi-level sociotechnical dynamics as an SR novelty claim.
- Delayed or ambiguous feedback as an SR novelty claim.
- Normal operation masking movement toward risk as an SR novelty claim.
- Boundary migration itself as an SR novelty claim.

These should be treated as established intellectual precursors or neighboring mechanisms in future SR manuscripts.

16. What Survives

- SR's use of drift across non-safety domains, provided each domain is operationalized rather than treated metaphorically.

- Visibility Lag as a candidate explicit measurable variable, though conceptually anticipated by delayed feedback.
- Cross-layer propagation ϕ as a candidate diagnostic extension, subject to comparison with network/interdependency science.
- Integration of drift diagnosis with restoration sufficiency and Recovery Slack.
- Prospective failure-surface mapping as a candidate operational application if it outperforms simpler boundary-migration models.

The survival is therefore narrower than 'SR discovered drift.'

**Surviving claim: SR operationalizes and couples drift across domains;
Rasmussen already explains the generic migration mechanism**

17. Revised Canonical Position of SR Drift

The canonical SR position should be revised to acknowledge that drift in the generic sense of gradual system migration under competing pressures has strong antecedents in Rasmussen and later safety science.

A defensible formulation is:

**SR Drift = domain-specific state deviation integrated with visibility,
propagation, and restoration diagnostics**

This formulation makes no claim that SR originated the general idea of drift into failure. Instead, SR's scientific burden becomes whether this integrated diagnostic architecture produces useful cross-domain measurement and prediction.

18. Status Table After Adversarial Test

Claim	Status	Reason
Gradual migration under pressure is uniquely SR	REJECTED	Rasmussen 1997 directly anticipates it.
Multi-level socio-technical drift is uniquely SR	REJECTED	Rasmussen explicitly spans government to operators.

Delayed recognition/feedback is uniquely SR	REJECTED as novelty	Core precursor exists in dynamic safety; SR may operationalize it.
Human adaptation is uniquely SR	REJECTED	Central to Rasmussen and resilience engineering.
SR drift can apply beyond safety boundaries	SURVIVES conceptually	Scope is broader, but must avoid vacuity.
ϕ propagation is an incremental SR contribution	UNRESOLVED	Needs comparison with network/interdependency models.
Visibility Lag adds predictive value	UNRESOLVED	Needs head-to-head empirical test.
SR drift + restoration predicts more than Rasmussen	UNRESOLVED	No prospective nested-model test yet.

19. Falsification Conditions

- If domain-specific SR Drift cannot be distinguished from ordinary trend/deviation measures, the construct should be narrowed or dropped in that domain.
- If Rasmussen-style pressure, boundary proximity, and adaptation variables predict future failure as well as SR's expanded model, SR has no demonstrated incremental value.
- If ϕ cannot be measured reproducibly or does not improve prediction beyond established network/interdependency metrics, it should not be treated as a distinct operator.
- If Visibility Lag is fully captured by conventional feedback delay and adds no predictive information, it should remain descriptive terminology rather than a separate diagnostic law.
- If restoration variables improve post-event management but not drift prediction, they should remain within SRD rather than be used to defend generic SR drift.

20. Discussion

The adversarial comparison substantially narrows SR's intellectual claim. Rasmussen anticipated far more of the generic drift architecture than a superficial comparison would suggest. His model already contains the central idea that ordinary, locally rational

adaptation under economic and workload pressure can migrate a socio-technical system toward an unsafe boundary while immediate feedback remains weak.

That is not a defeat for SR as an architecture, but it is a defeat for any claim that SR independently discovered the generic mechanism of systemic drift.

The comparison also clarifies where SR must now compete. Its value cannot come from saying that systems drift. It must come from whether SR can measure multiple kinds of drift, map their propagation, identify recognition lag, locate the next failure surface, and connect diagnosis to restoration more effectively than existing safety and systems methods.

SR value, if any = integration + operationalization + incremental prediction

This is a stronger scientific position because it converts provenance overlap into a testable boundary.

21. Conclusion

Rasmussen survives the challenge strongly. SR's generic drift mechanism does not.

Generic SR Drift $\approx$ strongly anticipated by Rasmussen-style migration

What remains is a narrower candidate contribution:

SR Drift* = measurable domain-state deviation + visibility + propagation + restoration context

That formulation is not yet empirically superior to dynamic safety theory. It is a hypothesis for further testing.

The next step should therefore not be another conceptual defense of drift. It should be a nested empirical comparison in a domain where both Rasmussen variables and SR variables can be measured prospectively.

If SR adds nothing beyond Rasmussen, SR must concede the overlap.

That is the result of this paper.

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